

Relativistic Electron Fluid Simulation and Studies on Electric Shock Wave Formation

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This study simulates the scenario when a relativistic electron beam enters high-density plasmas under the fast ignition condition. A relativistic fluid simulation code was developed for simulating relativistic electron transport in high dense plasmas. It is found that a shock wave is formed in a two temperature electron plasma. Namely an electron acoustic mode exists in the two component electron plasma. When relativistic electrons enter high-density plasmas at the relativistic electron expansion front, a shock wave front is formed in our simulation results. This shock wave in the simulation satisfies the relativistic Rankine–Hugoniot condition and Mach number condition.

KEYWORDS: relativistic electron transport, shock wave, fast ignition, electron acoustic wave
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1. Introduction

In 1994, a new ignition scheme was proposed by Tabak, which is based on the development of an ultra short pulse and ultra intense laser.¹⁾ This is called the “fast ignition scheme.” In the fast ignition scheme, relativistic electrons generated by the interaction of an ultra intense laser with high-density plasmas play an important role in igniting a thermo-nuclear burn. Here, the elucidation of the energy transport and relaxation process of relativistic electrons is an important subject. The related relativistic electron current is greater than several 100 MA, resulting in the complicated electromagnetic fluid phenomena, where the effect of the electric resistance of the return current becomes important. Computer simulations in the study of the above mentioned transport physics are indispensable. In recent years, relativistic electron kinetics has been studied mainly with particle in cell (PIC) codes using significantly scaled-down models.^{2–6)} However, large-scale space simulation is difficult by the PIC scheme in the present computer capability. The relativistic electron density of about 3 MeV is estimated to be $\sim 10^{21} \text{ cm}^{-3}$ for a typical condition. Thus, the computer memory size is not enough to calculate the actual number of electrons. In this paper, we simulate relativistic electron dynamics by using the relativistic fluid equation. We used the two fluid model of high- and low-temperature plasmas, and we analyzed those electromagnetic fluid equations by using the CIP scheme⁷⁾ instead of the PIC scheme.

2. Fluid Equations for Electron Dynamics

For cold electrons, we use a non-relativistic fluid equation as follows:

$$\frac{\partial n_c}{\partial t} + u_c \frac{\partial n_c}{\partial x} = -n_c \frac{\partial u_c}{\partial x} \quad (1)$$

$$\begin{aligned} \frac{\partial u_c}{\partial t} + u_c \frac{\partial u_c}{\partial x} = & -\frac{1}{m_e n_c} \frac{\partial (n_c T_c)}{\partial x} - \frac{e E_x}{m_e} \\ & + v_{iec}(u_i - u_c) - v_{eeh}(u_c - u_h) \end{aligned} \quad (2)$$

$$\begin{aligned} \frac{\partial T_c}{\partial t} + u_c \frac{\partial T_c}{\partial x} = & -\frac{2}{3} T_c \frac{\partial u_c}{\partial x} \\ & + \frac{2}{3} [-v_{iec} m_e (u_i - u_c)^2 + v_{eeh} \gamma m_e (u_c - u_h)^2] \end{aligned} \quad (3)$$

where n_c , u_c , T_c are number density, flow velocity, and temperature, m_e is the rest mass of an electron, e is the elementary charge. v_{iec} is collision frequency between the cold electrons and ions. Then v_{eeh} is the collision frequency between the cold and hot electrons. γ is the Lorentz factor for hot electron. u_h is flow velocity for hot electrons, u_i is the flow velocity for ions. E_x is electric field of hot electron beam propagation direction.

For hot electrons, we have to use the relativistic equations of fluid dynamics.⁸⁾ This relativistic equation is as follows:

$$\frac{\partial T^{ik}}{\partial x^k} = 0 \quad (4)$$

$$T^{ik} = \frac{1}{c^2} (\rho_h c^2 + P_h + \rho_h e_{sie}) u^i u^k + P_h \eta^{ik} \quad (5)$$

where T^{ik} is the energy-momentum tensor, c is the speed of light, u^i is the fluid four-velocity, $\eta^{ik} = \text{diag}(-1, 1, 1, 1)$, P_h , ρ_h , and e_{sie} are pressure, mass density, and specific internal energy per unit volume, respectively. Assuming isotropy, the energy-momentum tensor is as follows:

$$T^{\mu\nu} = \begin{pmatrix} \rho_h c^2 + \rho_h e_{sie} & 0 & 0 & 0 \\ 0 & p_h & 0 & 0 \\ 0 & 0 & p_h & 0 \\ 0 & 0 & 0 & p_h \end{pmatrix} \quad (6)$$

Then, the equation of continuity is

$$\frac{\partial (\gamma \rho_h)}{\partial x^i} = 0 \quad (7)$$

Moreover, the equation of state is

$$P_h = (\gamma_h - 1) \rho_h e_{sie} \quad (8)$$

where γ_h is ratio of specific heat.

Finally, the equation for hot electrons is given as follows:

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$$\frac{\partial(\gamma\rho_h)}{\partial t} + u_h \frac{\partial(\gamma\rho_h)}{\partial x} = -\gamma\rho_h \frac{\partial u_h}{\partial x} \quad (9)$$

$$\frac{\partial u_h}{\partial t} + u_h \frac{\partial u_h}{\partial x} = -\frac{u_h}{H} \frac{\partial P_h}{\partial t} - \frac{c^2}{H} \frac{\partial P_h}{\partial x} + \frac{1}{\rho_h} \{-en_h E_x + v_{ieh}\gamma m_e n_h (u_i - u_h) + v_{eeh}\gamma m_e n_h (u_c - u_h)\} \quad (10)$$

$$\frac{\partial H}{\partial t} + u_h \frac{\partial H}{\partial x} = -H \frac{\partial u_h}{\partial x} + \frac{\partial p_h}{\partial t} - v_{ieh}\gamma m_e n_h (u_i - u_h)^2 - v_{eeh}\gamma m_e n_h (u_c - u_h)^2 \quad (11)$$

where n_h is the number density of hot electrons and H is the relativistic enthalpy. H is defined as

$$H \equiv \gamma^2(\rho c^2 + \rho e + p) \quad (12)$$

Now, we believe that two fluid plasmas interact only with an electric field. The electric field E_x is calculated using the Ampere–Maxwell equation. For this equation, elementary charge is considered to be 1/100 its exact value to solve a diffusion phenomenon in large scale and in a long time step.⁹⁾ Namely, we calculate the Poisson equation as follows:

$$\frac{\partial E_x}{\partial t} = 4\pi\varepsilon(n_h u_h + n_c u_c - n_i u_i) \quad (13)$$

where ε is 1/100 the elementary charge. n_{eh} , u_{eh} are the number density and the flow velocity of hot electrons, n_{ec} , u_{ec} are the number density and the flow velocity of cold electrons, n_i , is number density of ions.

3. Dispersion Relation for an Electrostatic Wave in a Two Component Electron Fluid

A background ion's motion was not considered here. The dispersion relation for two component electron plasmas is calculated as follows:

$$\begin{aligned} \frac{\omega^2}{\omega_p^2} = & \frac{1}{2} \cdot \frac{1 + 3k^2(T_{c0} + T_{h0})}{m_e \omega_p^2} \\ & \pm \left[\frac{1 + 6k^2(T_{c0} + T_{h0})}{m_e \omega_p^2} + \frac{9k^4(T_{c0} + T_{h0})^2}{m_e^2 \omega_p^4} - \frac{36k^4 T_{c0} T_{h0}}{m_e^2 \omega_p^4} - \frac{12k^2(T_{c0} \omega_{ph}^2 + T_{h0} \omega_{pc}^2)}{m_e \omega_p^4} \right] \end{aligned} \quad (14)$$

This dispersion relation not only includes the electron plasma wave mode, but also the electron acoustic mode. In Fig. 1, we show the dispersion relation for two component electron plasmas. In the small wave number region, a low frequency mode corresponds to the acoustic mode, whose frequency is proportional to the wave number in a manner similar to an ion acoustic wave. This indicates that an electron shock wave can be formed in a two electron component plasma.

4. Simulation Results

First, we consider collision less case. As an initial condition, the relativistic electron density was set up as in Fig. 2. The ion density is uniformly set to 10^{23} cm^{-3} . The hot electron number density is set to 10^{20} cm^{-3} and the cold electron density is set in accordance with the charge neutrality condition. The initial two fluid velocities are set to zero, following which the relativistic and cold electron temperatures are 100 and 10 keV, respectively.

The spatial profiles of the relativistic electron density, velocity, and pressure at $t = 0.6 \text{ ps}$ are shown in Fig. 3. In Fig. 3, we found that the relativistic electron forms shape like a shock wave and transport in cold electrons. Based on the system that rode on the shock like front, we can obtain the relativistic Rankine–Hugoniot relation. We can calculate the propagation speed of a shock wave front by the relativistic Rankine–Hugoniot relation.

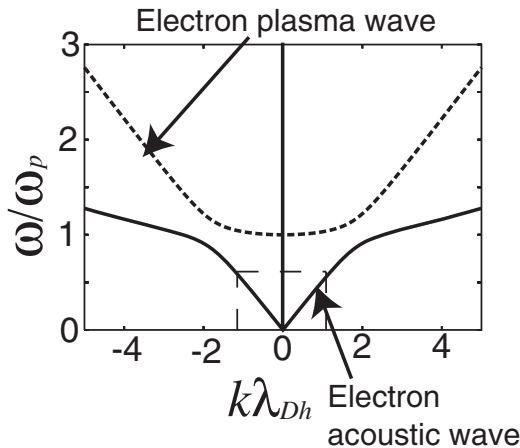


Fig. 1. Dispersion relation for two component electron plasmas.

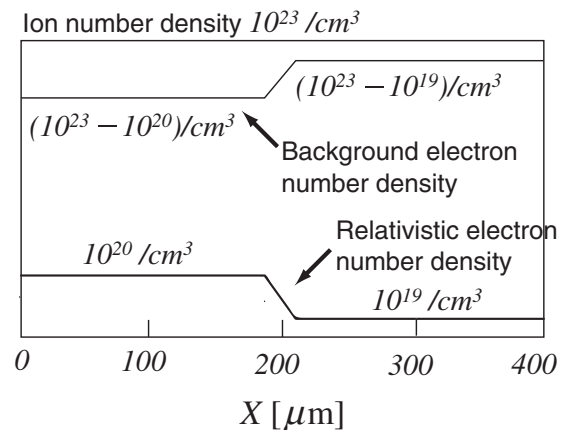


Fig. 2. Initial condition of the hot and cold electron density profile.

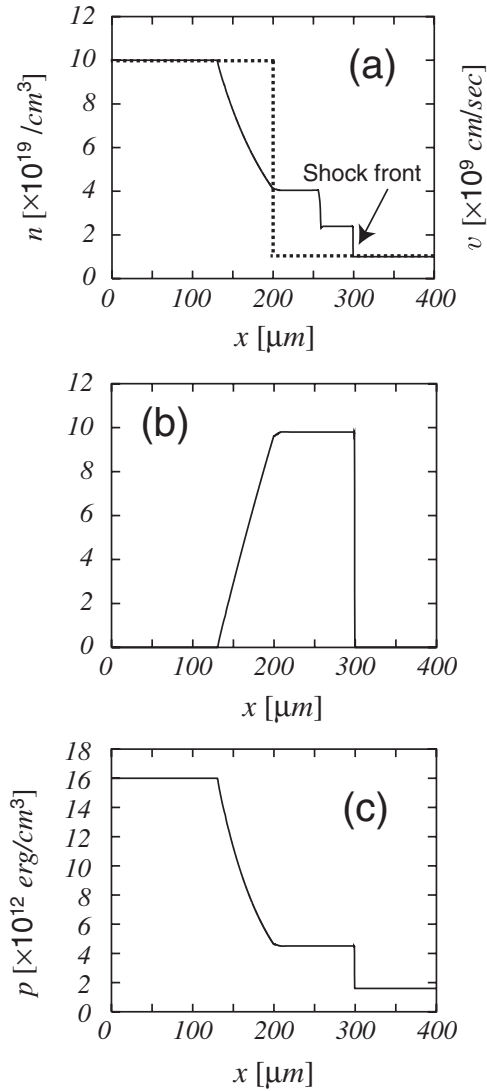


Fig. 3. The hot electron is formed in the shock wave front, we call the “electron acoustic wave,” in high-density plasmas: (a) Number density for the relativistic electron: at initial condition (dotted line) and at $t = 0.6 \text{ ps}$ (solid line). (b) Velocity for the relativistic electron, at $t = 0.6 \text{ ps}$ and (c) pressure for the relativistic electron, at $t = 0.6 \text{ ps}$.

$$V_s = -\frac{\gamma_1 \rho_1 u_1 - \gamma_2 \rho_2 u_2}{\gamma_1 \rho_1 - \gamma_2 \rho_2} = 1.71 \times 10^{10} \text{ cm/s}$$

where V_s is the propagation speed of the shock wave front, and subscripts 1 and 2 show the forward and backward values of the shock front, respectively. The speed of the shock like front is about $1.65 \times 10^{10} \text{ cm/s}$ in our simulation; this speed correspond with result of relativistic Rankine–Hugoniot relation. In addition, the Mach number is 0.95 at the backward point of the shock like front and 1.03 at the forward point of the shock like front. Therefore, we believe that the relativistic electron forms a shock wave; we call it the “electron acoustic wave.”

Next, we consider the case that included a collision term. We consider the case where the drift speed of a relativistic electron is given. Now, the drift velocity of the relativistic electron is set to $0.3c$, where c is the speed of light. The remaining conditions are the same as a previous case. In Fig. 4, we show the simulation results. In these results, the

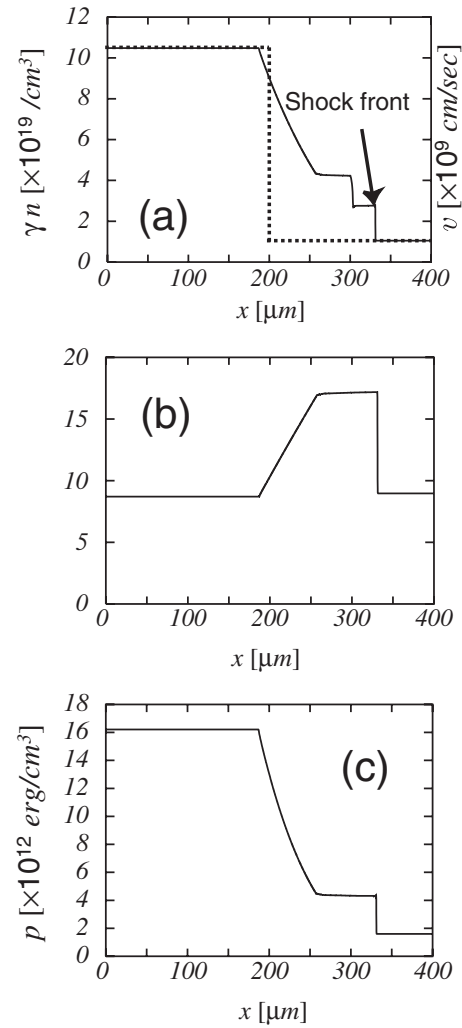


Fig. 4. The shock wave is formed as in the case of the relativistic electron with an initial drift velocity of $0.3c$. (a) Number density for the relativistic electron: at initial condition (dotted line), at $t = 0.6 \text{ ps}$ (solid line). (b) Velocity for the relativistic electron, at $t = 0.6 \text{ ps}$ and (c) pressure for the relativistic electron, at $t = 0.6 \text{ ps}$.

relativistic electron assumes a form similar to a shock wave. The speed of the shock front is $2.19 \times 10^{10} \text{ cm/s}$, and this velocity exceeds the initial drift velocity. Also, this speed corresponds to that obtained by the relativistic Rankine–Hugoniot relation.

$$V_s = -\frac{\gamma_1 \rho_1 u_1 - \gamma_2 \rho_2 u_2}{\gamma_1 \rho_1 - \gamma_2 \rho_2} = 2.2 \times 10^{10} \text{ cm/s}$$

5. Conclusion

We developed 1D two component electron plasma fluid code using the Mason Model and calculated the relativistic electron transport in high-density plasmas. It was shown that a possibility existed for the electron to have a sound wave mode from a dispersion relation of a two component electron fluid plasma. The simulation result clarified that a relativistic electron formed a shock wave (“electron acoustic wave”) in two component electron plasmas. It is found that considering an elementary charge of $1/100$ its exact value, the simulation was carried out for a longer time step and larger space.

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